

NIEL ABHISHEK J DAVID

+91 6360701051 • nielabhishek6265852@gmail.com • Bangalore 560058

• [LinkedIn](#) • [GitHub](#) • [PortFolio](#)

PROFESSIONAL SUMMARY

Aspiring Computer Science Engineer with experience building backend applications using Java, Spring Boot, Python, FastAPI, and SQL. Skilled in designing REST APIs, cloud-native applications, and scalable software systems with hands-on experience in AWS, Docker, CI/CD, and database design. Passionate about backend engineering, distributed systems, and building reliable software.

EXPERIENCE

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- ❖ **Software Development Intern | Indian Institute of Science** *Jan 2025 – Jun 2025*
 - Developed backend services using FastAPI, Firebase, and WebSockets for an AI-powered Smart Mirror deployed on NVIDIA Jetson Nano.
 - Designed and implemented the Digital Wardrobe module with cloud storage, authentication, and real-time synchronization. Integrated computer vision and LLM-based features for outfit analysis, face recognition, and personalized recommendations.
 - Collaborated on full-stack development using React, Firebase, and REST APIs to build an interactive smart mirror experience.

 - ❖ **Data Analyst (GAT) | Volvo Group India Pvt Ltd** *March 2026 – Present*
 - Engineered Python (Pandas/NumPy) and SQL data wrangling scripts on Azure Databricks to automate data cleansing and preprocessing for aftermarket planning pipelines, reducing processing times.
 - Designed automated ETL workflows using Power Automate and Azure services to streamline cross-system data flows between Dataverse and enterprise storage.
 - Developed interactive Power BI dashboards and automated reports connected to SQL databases and Dataverse/SharePoint, transforming raw metrics into key business KPIs.

PROJECTS

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- ❖ **FitTrack Pro**
Tech Stack: Java 17, Spring Boot 3, Spring Security, JWT, Hibernate, MySQL, React.js, Docker, Swagger, MapStruct
 - Architected and developed a production-ready fitness management platform featuring secure authentication, role-based authorization, workout scheduling, exercise management, progress analytics, and personalized dashboards using Spring Boot and React.
 - Designed scalable RESTful APIs with layered architecture, Hibernate/JPA, DTO mapping, validation, global exception handling, and optimized database relationships following SOLID design principles.
 - Integrated AI-powered workout generation, official Spotify Web API connectivity, responsive analytics dashboards, Dockerized deployment, Swagger API documentation, and automated testing to deliver a modern SaaS-style fitness application.

 - ❖ **Digital Wardrobe**
Tech Stack: FastAPI, React.js, WebSockets, Gemini LLM, OpenCV, Face-Recognition, Firebase(Database)
 - Architected AI-driven microservices using FastAPI and WebSockets to deliver low-latency communication between smart mirror services.
 - Implemented secure facial recognition, Firebase Authentication, and cloud-based wardrobe management using Firestore and Firebase Storage.
 - Containerized backend services with Docker and integrated Gemini LLM for personalized outfit recommendations and voice-assisted interactions.
 - Designed scalable backend services using modular architecture and REST APIs to support real-time communication across distributed components.

❖ **Fashion Recommendation System**

Tech Stack: TensorFlow, EfficientNetB3, Flask, SQLAlchemy, MLflow, React

- Developed an EfficientNetB3-based deep learning model achieving 92% accuracy for personalized fashion recommendations.
- Designed PostgreSQL schemas with SQLAlchemy for efficient storage of user profiles, measurements, and recommendation history.
- Containerized ML inference services using Docker and automated model validation and deployment workflows with GitHub Actions.

❖ **Cloud-Based Attendance System**

Tech Stack: React.js, Node.js, Firebase Auth/Cloud firestore, CI/CD, GitHub Actions

- Built a responsive school attendance system with secure login, real-time updates via Firebase Firestore, and student profile management.
- Implemented a modular backend with RBAC authentication and analytics dashboard to visualize attendance patterns.
- Cloud-native solution automating attendance tracking with secure access and real-time updates.

❖ **Network Security Firewall Solution**

Tech Stack: Python, Tornado, Flask, Distributed Caching, Intrusion Detection Systems

- Developed a rule-based intrusion detection system using Python, Flask, and Tornado to support concurrent traffic inspection and centralized security rule management.
- Built RESTful management APIs allowing administrators to dynamically configure security policies and monitor suspicious network activity.
- Optimized session management using Redis caching and containerized the entire application with Docker for portable, high-throughput deployment.

EDUCATION

❖ **B.Tech in Computer Science Engineering**

PES University, RR Campus, Bengaluru

Sept 2022 – Aug 2025

CGPA – 7.73

TECHNICAL SKILLS

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- **Programming Languages:** Python, Java, C#, JavaScript, Bash
 - **Frontend Development:** HTML, CSS, React.js
 - **Backend Development:** Servlets, Spring IoC & MVC, Spring Boot, Spring Security, Hibernate, JPA, FastAPI, Django, REST APIs, JWT Authentication, ASP.NET Core
 - **Database:** MySQL, MongoDB, Oracle SQL(PL-SQL)
 - **Machine Learning Libraries:** Pandas, NumPy, TensorFlow, Scikit-Learn
 - **Testing & API Tools:** JUnit 5, Mockito, Swagger/OpenAPI, Postman
 - **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Docker Compose, GitHub Actions, CI/CD
 - **Developer Tools:** Git, GitHub, Maven, Linux, IntelliJ IDEA, Eclipse, VS Code

CERTIFICATIONS

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- Beginner's guide to Linux Kernel Development. ([Certificate](#)) _
 - AWS educate Introduction to Cloud. ([Storage](#), [Compute](#), [Serverless](#), [Networking](#))
 - Jira Work Management Fundamentals Badge ([Atlassian University](#))
 - JSpiders Java Full Stack + GenAI Course